Taylor Dinkins

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Research Interests

My research interests focus on machine learning and deep learning methods for detecting and forecasting anomalies and events, with an emphasis on real-world usability. I am particularly interested in developing and applying machine learning to high-impact societal domains, such as AI safety and malicious use detection and climate change. More broad interests include time series forecasting, representation learning, and reinforcement learning for applied problems.

Education

Oregon State University September 2026

Ph.D. in Computer Science
Thesis: *Machine Learning Algorithms for Anomaly Detection and Event Detection*
Advisor: Weng-Keen Wong

California State University Channel Islands May 2017

B.S. in Computer Science & B.S. in Mathematics, *summa cum laude*

Publications

Peer-Reviewed

Dinkins, T., Wong, W.K., Amogi, B., Pesantez-Cabrera, P., Patel, J., Khot, L., and Fern A. (2027). Improving spatial generalization with edge-gated spatiotemporal graph neural networks for temperature inversion forecasting. *Innovative Applications of Artificial Intelligence (IAAI), 2027.*

Dinkins, T., Wong, W.K., Amogi, B., Pesantez-Cabrera, P., Patel, J., Khot, L., and Fern A. (2026). Localized near surface temperature inversion forecasting using long short-term memory. *Innovative Applications of Artificial Intelligence (IAAI), 2026.*

Dinkins, T., Wong, W.K., Chen, H., Liu, Y., and Ishikawa, K. (2025). Incorporating interactive user feedback through precision matrix adjustments for high-dimensional anomaly detection. *IEEE International Conference on Data Mining (ICDM), 2025.*

Dinkins, T., Bhattacharyya, S., Chatterjee, S., Reis, S., and Wong, W.K. (2022). Towards explainable precision changepoint detection through linear decomposition. *Anomaly and Novelty Detection, Explanation and Accommodation Workshop (ANDEA), 2022.*

In Review

Dinkins, T., Wong, W.K., Bhattacharyya, S., Chatterjee, S., and Raffa, G. (2025). Online precision matrix changepoint detection with localization to groups of dimensions. *Data Mining and Knowledge Discovery (DMKD) Journal* (post-revision).

Teaching

Instructor of Record, **Oregon State University**

CS 325 Analysis of Algorithms	Summer 2019

Teaching Assistant, **Oregon State University**

CS 534 Machine Learning	Fall 2019, Fall 2020
CS 434 Machine Learning & Data Mining	Spring 2020
CS 535 Deep Learning	Winter 2020
CS 536 Probabilistic Graphical Models	Winter 2020
CS 519 Data Science Tools & Programming	Winter 2019
CS 461 Senior Software Engineering Project	Fall 2018

Experience

Research Assistant, **Oregon State University**	January 2021–September 2026
ML Research Intern, **Lawrence Livermore National Laboratory**	June 2022–December 2022
ML Engineer Intern, **Syntiant Corporation**	June 2021–September 2021
Data Specialist, **Ocean View School District**	August 2017–June 2018
Undergraduate Research Intern, **CERN**	June 2016–August 2016
Information Technology Assistant, **CSU Channel Islands**	August 2013–August 2017

Presentations

Oregon AI Research Workshop (Poster) April 2026
Localized near surface temperature inversion forecasting using long short-term memory

IEEE International Conference on Data Mining November 2025
Incorporating interactive user feedback through precision matrix adjustments for high-dimensional anomaly detection

Oregon State University, AI Week (Poster) April 2024
Online precision matrix changepoint detection with localization to groups of dimensions

PPI Center Industry Advisory Board (Poster) April 2024
Online precision matrix changepoint detection with localization to groups of dimensions

Anomaly and Novelty Detection, Explanation and Accommodation Workshop August 2022
Towards explainable precision changepoint detection through linear decomposition

Honors and Awards

Program Honors Mathematics, **CSU Channel Islands**	May 2017
NSF IRES Award, **National Science Foundation via Yongsheng Gao**	June 2016
Dean's List (7) & President's List (3), **CSU Channel Islands**	August 2012–May 2017

Skills

Programming: Python, SQL, Julia
Frameworks & Libraries: PyTorch, scikit-learn, NumPy, pandas, statsmodels
Tools: Git, Linux, LaTeX, Jupyter, Matplotlib

References

Dr. Weng-Keen Wong
Professor, School of Electrical Engineering and Computer Science
Oregon State University
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Dr. Sean McGregor
Fellow, Berkman Klein Center for Internet & Society at Harvard University
Cofounder, AI Verification and Evaluation Research Institute (AVERI)
smcgregor@cyber.harvard.edu

Dr. Sharmodeep Bhattacharyya
Associate Professor, Department of Statistics
Oregon State University
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